DO YOU KNOW?

A **carbon sink** is a natural or artificial reservoir that accumulates and stores carbon for an indefinite period. The process by which carbon sinks remove carbon dioxide (CO_2) from the atmosphere is known as **carbon sequestration**. The main **natural carbon sinks** are **plants**, the **ocean** and **soil.**

7.4 Biomes

An ecosystem consists of a biological community and an abiotic environment. Ecosystem may be broadly divided into land or *terrestrial ecosystem* and water or *aquatic ecosystem*. The aquatic ecosystem can be further divided into freshwater and marine ecosystem.

An ecosystem becomes a biome when it extends over a large area. According to **I.G. Simmons** (1982) the most extensive ecosystem unit which is convenient to designate is called a '*Biome*'. It may be concluded that a biome is in fact a large ecosystem where we study the total assemblage of plant and animal communities. Since vegetation is the most dominant component of a biome and as vegetation and climate are very intimately related, the world is divided into a number of biomes based on major world climatic types (Figure 7.7).

7.4.1. Types of Biomes

World Biomes are mega ecosystems existing and operating over large areas. These divisions are based on climate pattern, soil types, and the animals and plants that inhabit an area. Basically, biomes are classified into two major groups such as *Aquatic biomes* and *Terrestrial biomes*.

Wetlands are transition zones between aquatic and terrestrial biomes

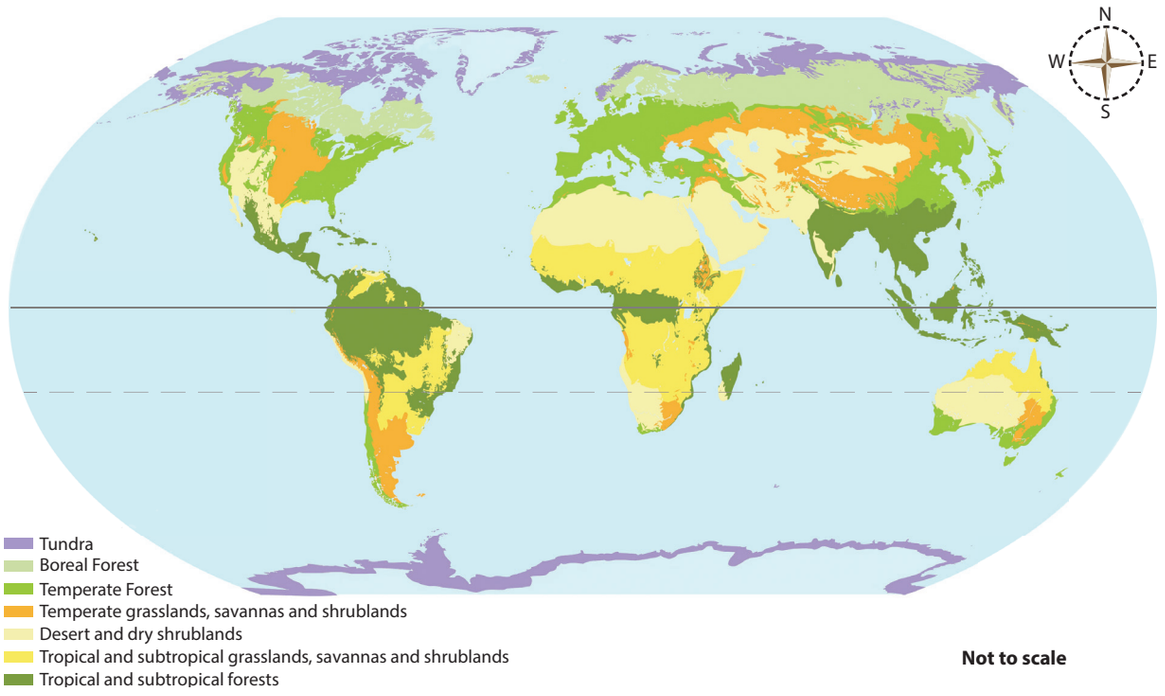


Figure 7.7 Biomes of the World

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